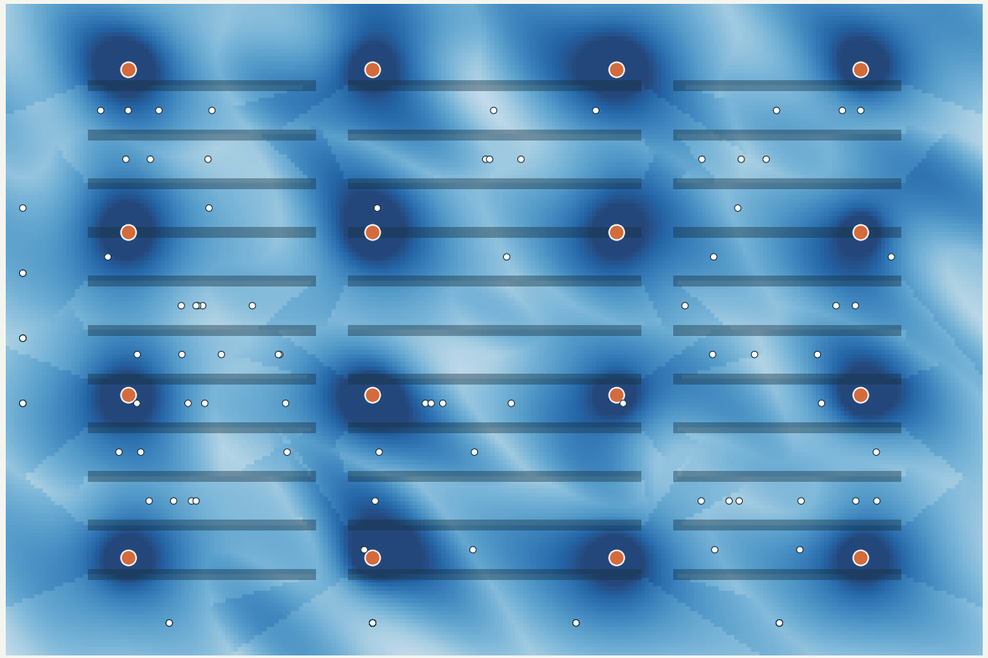
RadioMap + WirelessJEV: coordinated warehouse connectivity

One system for site learning, mission-aware decisions, packet recovery and operator visibility.

The test warehouse · 100 robots / 16 AP sites



LEARN THE SITE

↓

COMPILE THE MISSION

↓

RECOVER THROUGH ANOTHER AP

Simulation evidence · hardware proof pending

INSIGHT **A comprehensive feedback architecture connects radio state, intent, decisions and outcomes.**

A robot needs commands on time—not just a fast link

A 50 ms command deadline makes short outages more important than average throughput.

CONTROL
600 B · 20 Hz
50 ms deadline

VIDEO
150 ms modelled deadline
On-time delivery matters

BULK
Buffered application data
Rate is not continuity

THE REAL COST
Command silence
A robot may stop

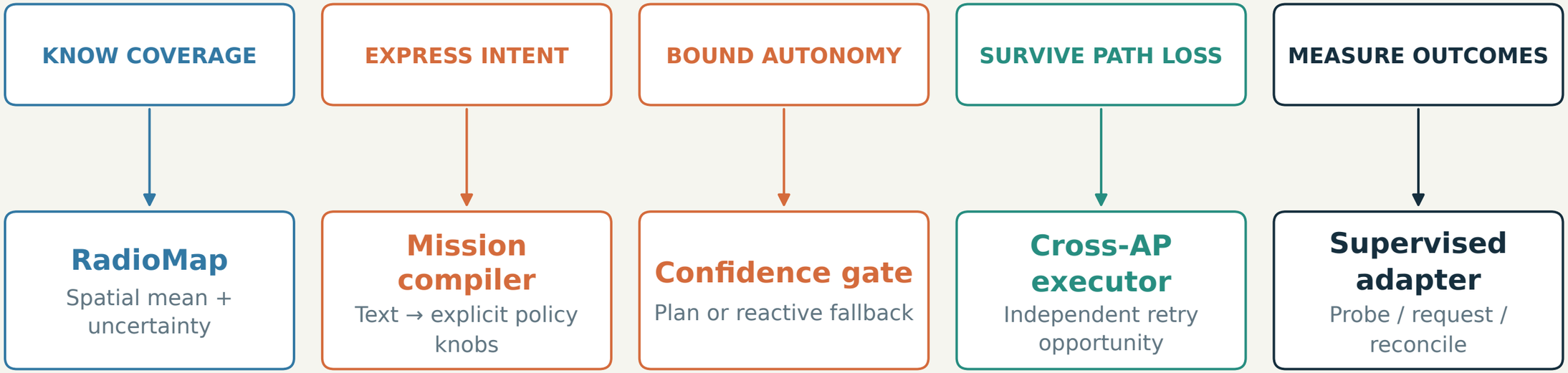


Average Mbps can look healthy while short commands expire.

INSIGHT **Reliability is a path-continuity problem before it is a peak-rate problem.**

Each system requirement has a corresponding control mechanism

Observability, intent, bounded autonomy, path continuity and measurement work together.



Five complementary responsibilities in one observable control system.

INSIGHT **Comprehensiveness means covering distinct failure modes—not adding unrelated features.**

RadioMap estimates; WirelessJEV decides; the executor recovers

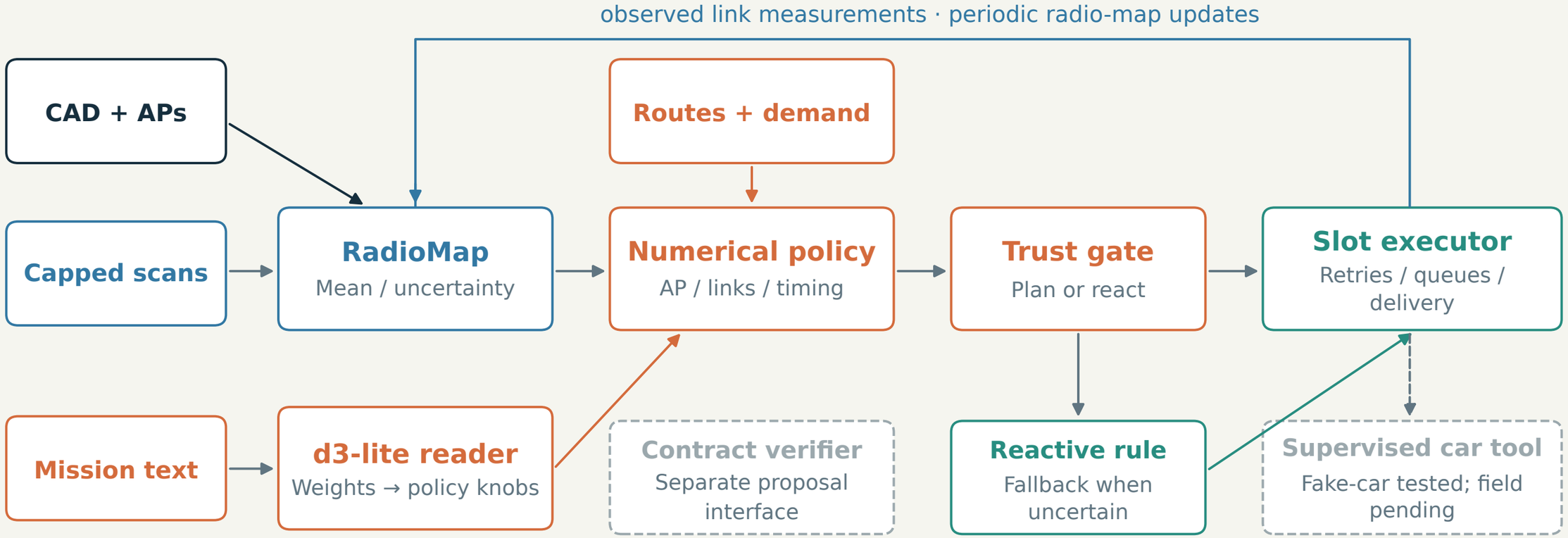
Separate spatial knowledge, mission-aware decisions and time-critical delivery.



INSIGHT **Clear interfaces keep uncertainty, decisions and physical outcomes from being confused.**

The system learns, plans, gates and closes the loop

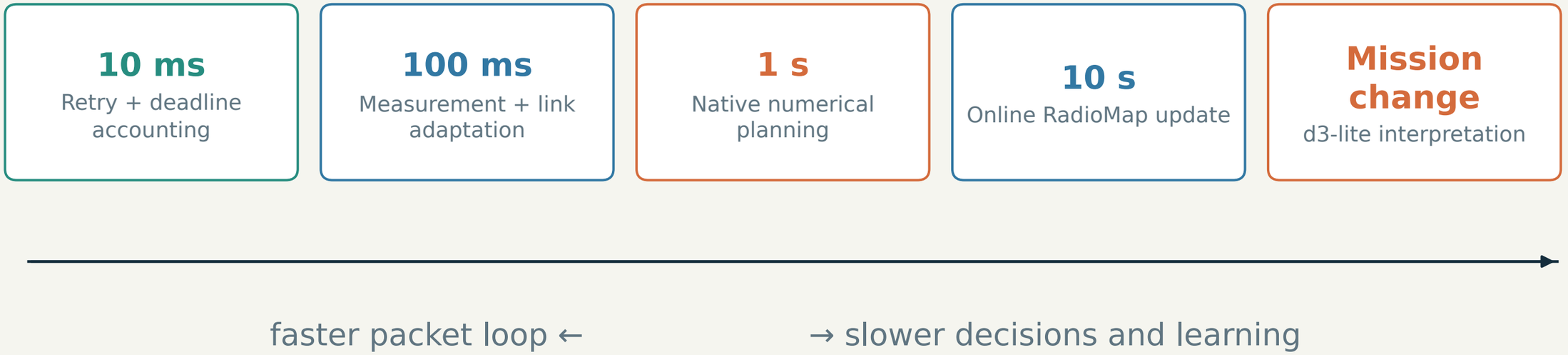
One connected simulated control path, with explicit boundaries around separate tools and hardware.



INSIGHT **The map’s uncertainty controls where planning is allowed to act.**

Separate fast packet recovery from slower mission planning

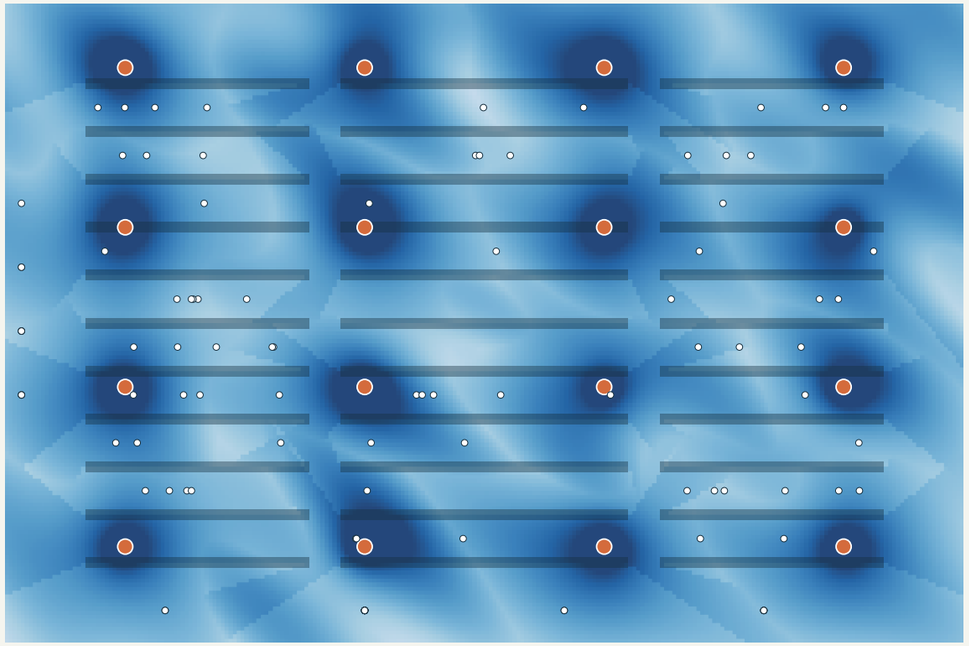
The simulator runs at 10 ms; measurements, planning and map updates run at different cadences.



INSIGHT **Language interpretation belongs outside the per-packet path.**

Every policy sees the same hidden warehouse conditions

100 robots, 16 AP sites and paired scenes with identical routes, fading and blockage.



SHARED
Routes · scene seed · fading · blockage

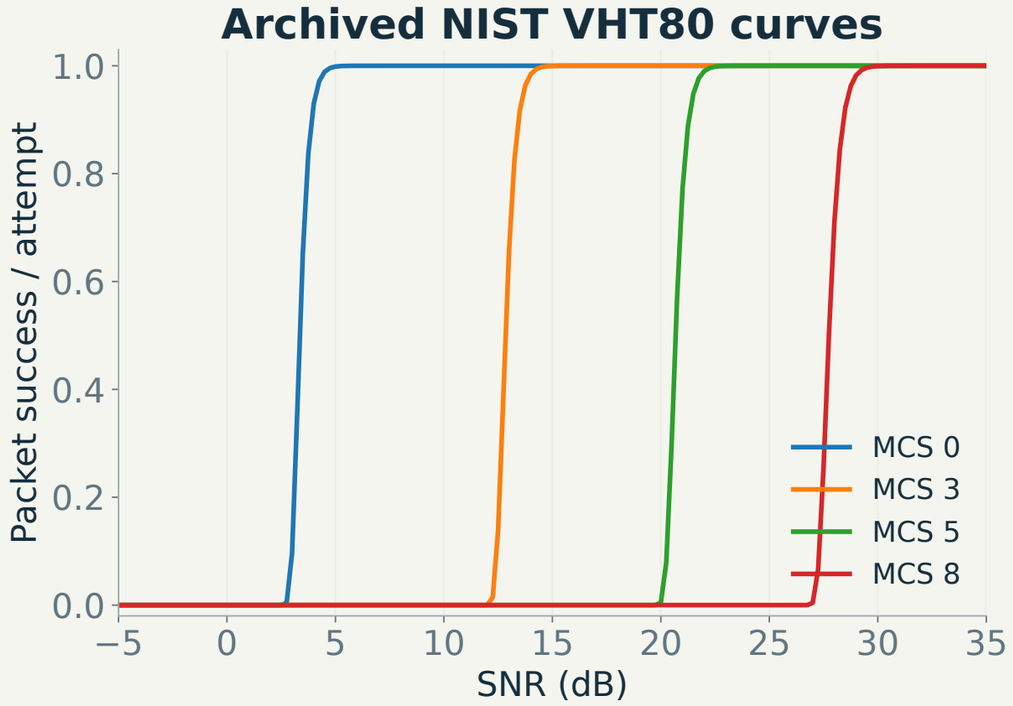
OBSERVED
Capped scans · lagged link measurements

HIDDEN
Instantaneous fades and blockage

INSIGHT **Shared scenes isolate policy differences within the model—not reality error.**

Standard link curves anchor a simplified slot model

Archived ns-3 NIST curves feed retries; Sionna is used in a separate coded-modulation study.



SOURCE
VHT80 / HT20 · 2 streams · 1,200 B

↓

EXTRAPOLATION
6 GHz: double 5 GHz rate at 160 MHz

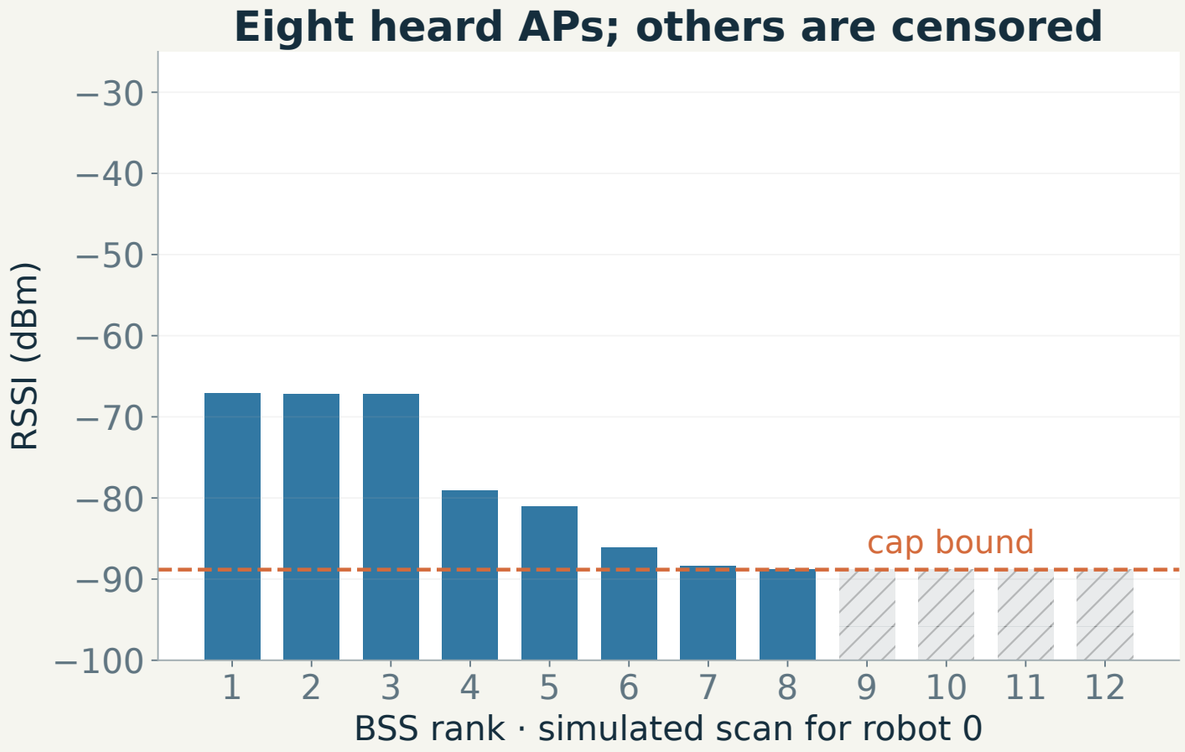
↓

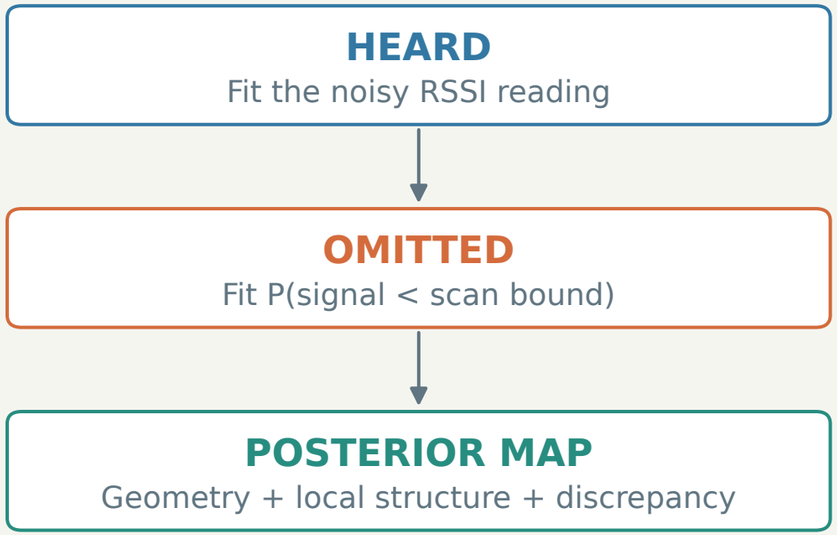
SYSTEM MODEL
10 ms slots · retry chains · simplified MAC

INSIGHT **This cross-layer Python simulator is not full ns-3 Wi-Fi or a 3GPP NR stack.**

A capped scan is evidence—not a list of invented measurements

Heard APs supply RSSI; omitted APs supply an inequality bounded by the scan cap.

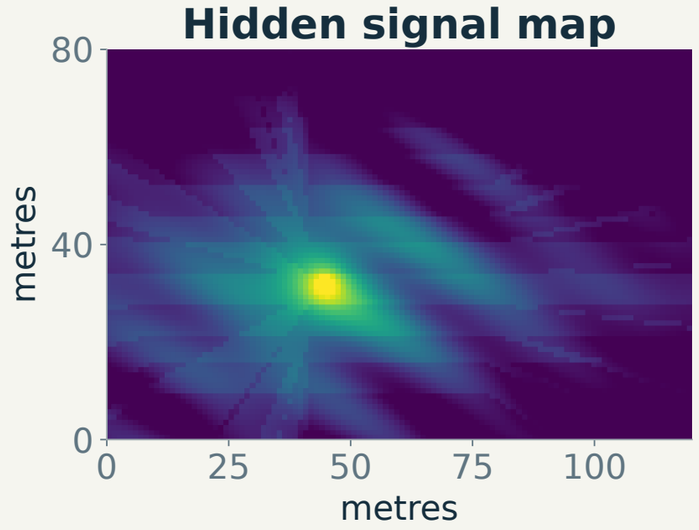


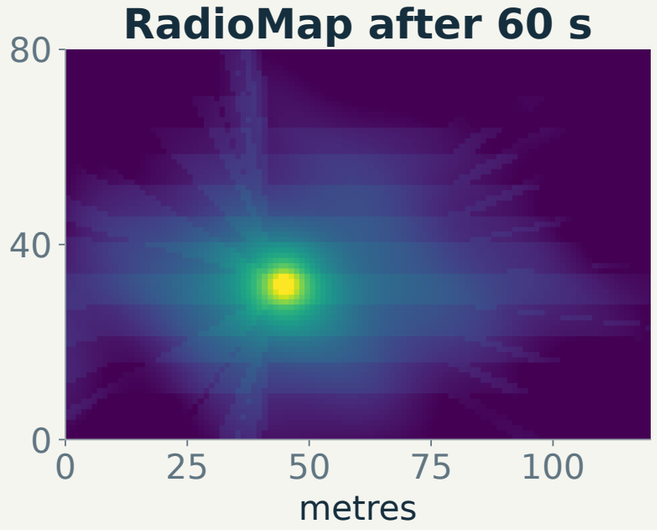


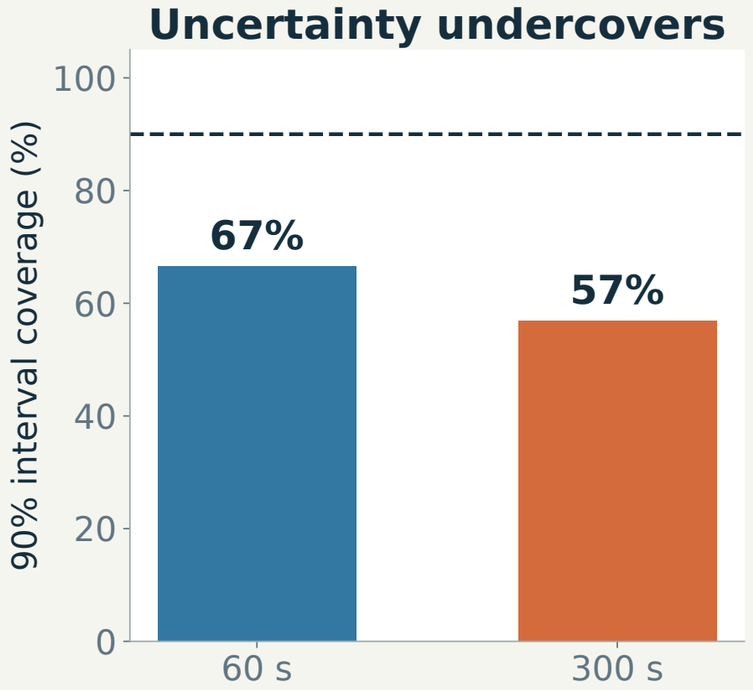
INSIGHT **Correct censoring is essential before trusting any learned radio map.**

One minute learns useful structure—but uncertainty undercovers

RadioMap reaches about 3.8 dB RMSE after 60 s in the six-site campaign.





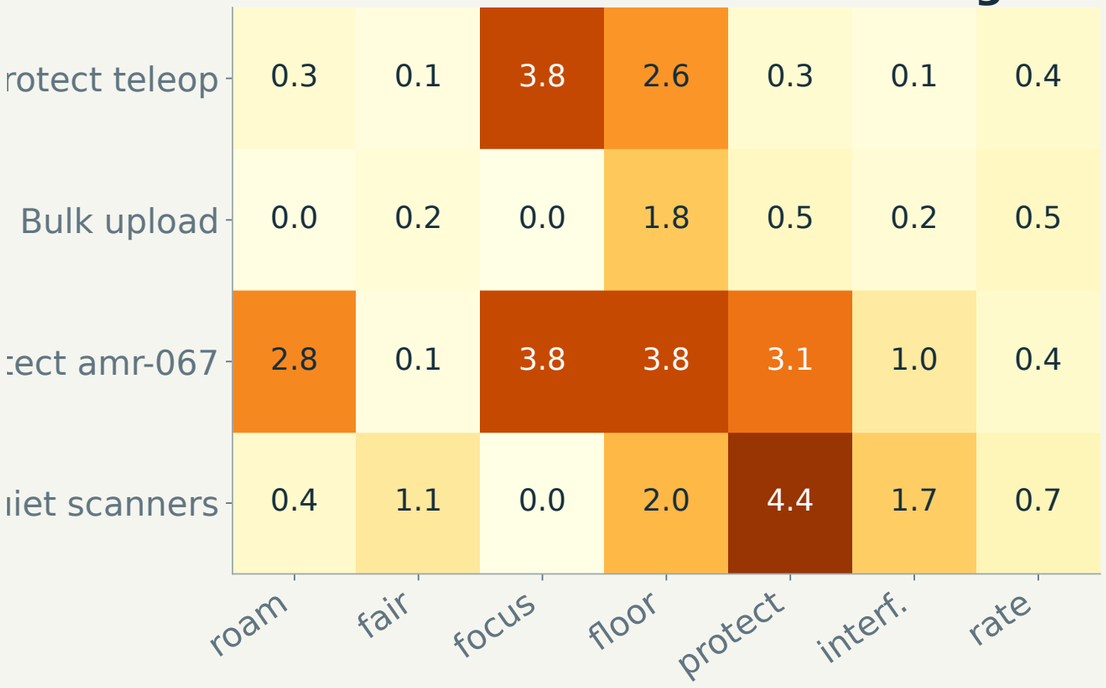


INSIGHT **More data improves mean error while structural uncertainty can become overconfident.**

Mission text becomes explicit policy knobs

Actual d3-lite readings control named-robot focus, roam caution, fairness and channel preference.

Actual d3-lite mission readings



TEXT → WEIGHTS
Inference on mission change

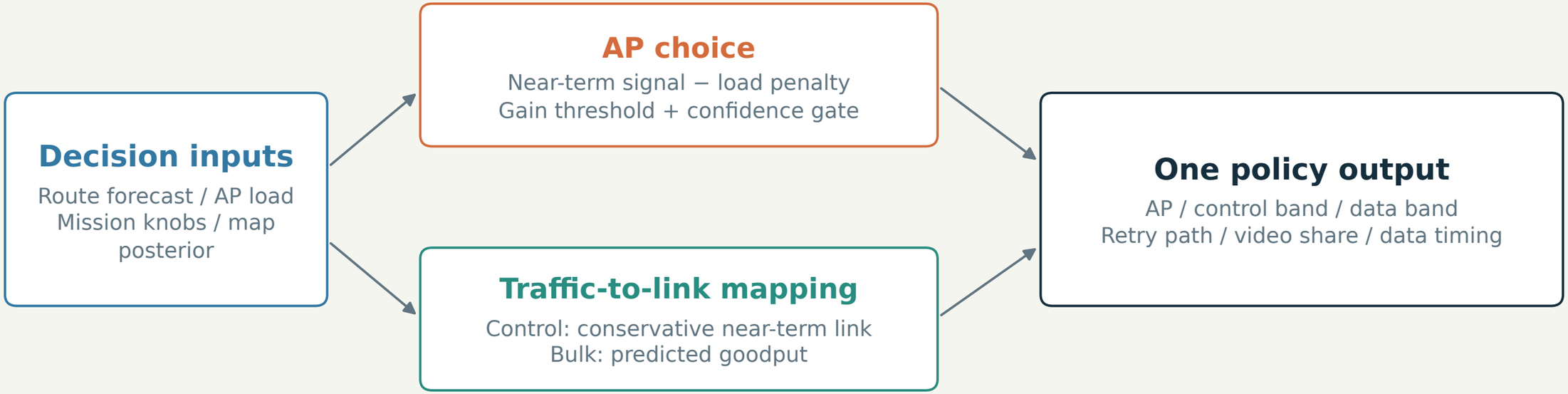
WEIGHTS → KNOBS
Thresholds / video share / focus

SEPARATE CONTRACT TOOL
Explicit groups / floors / proposal checks

INSIGHT **The native mission adapter is a weight compiler; the richer contract tool is separate.**

One planning step chooses AP, control link and data behavior

The numerical policy combines near-term map forecasts, load and operator-controlled thresholds.

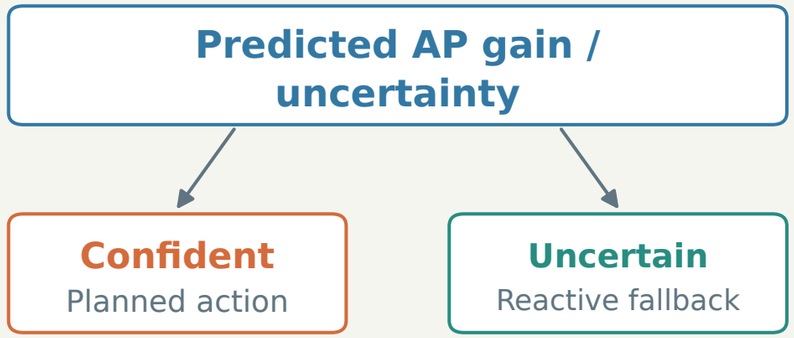


Explicit numerical heuristics, evaluated once per second—not a global optimum claim.

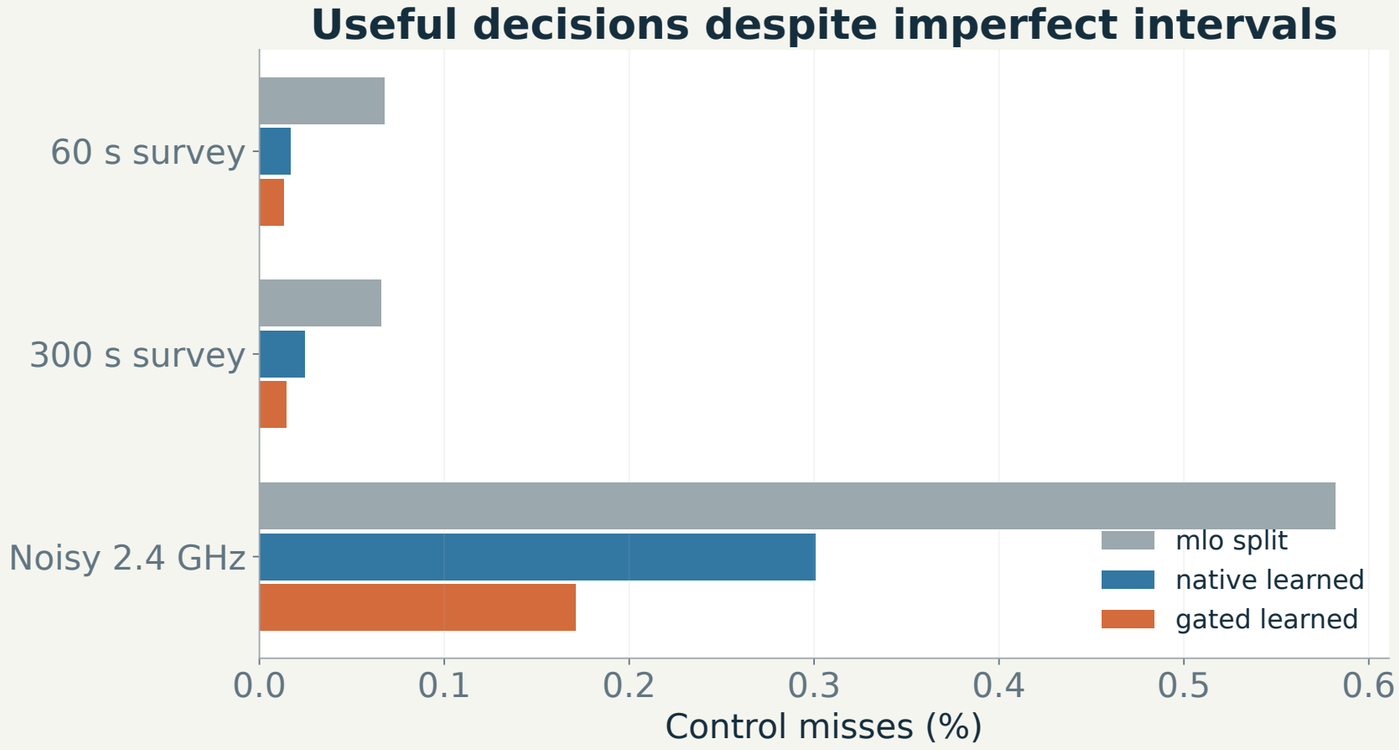
INSIGHT **Different traffic classes need different decisions over the same physical links.**

A confidence gate prevents uncertain maps from steering freely

Compare predicted gain with map uncertainty; use the reactive rule when confidence is weak.



$$z = \Delta / \sqrt{\sigma_{old}^2 + \sigma_{new}^2}$$



INSIGHT **The gate is a decision safeguard, not a certificate of calibrated safety.**

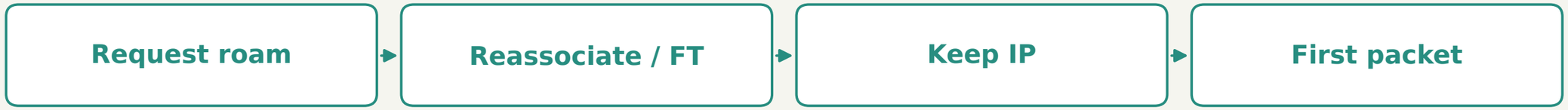
A teardown and a within-connection roam are different paths

The car tools can probe and time both paths; the latency advantage is still assumed in simulation.

NetworkManager profile activation



Within-connection roam target



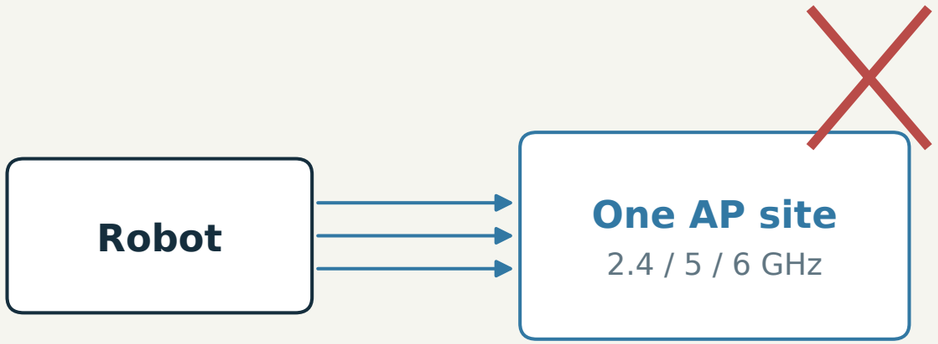
Built: read-only probes + stage clock checks + supervised request. Unmeasured: real interruption.

INSIGHT **Measure the interruption before claiming a fast-roam result on hardware.**

MLO protects bands; a second AP path protects attachment

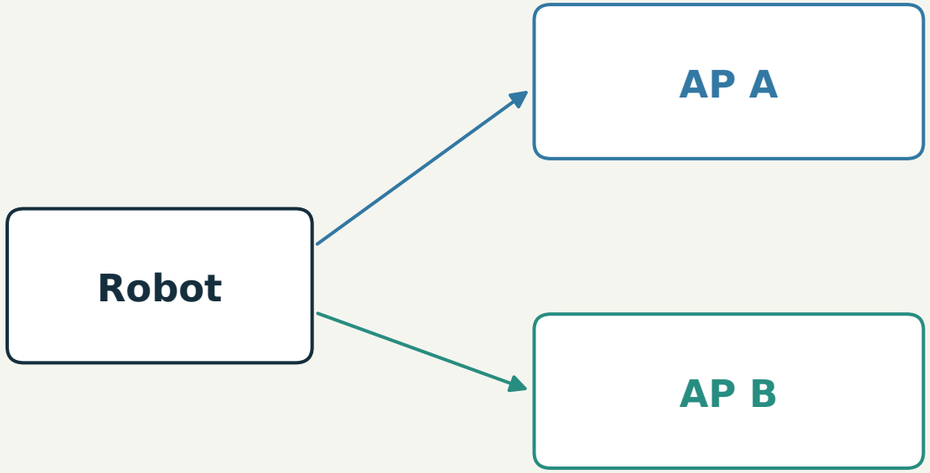
A blockage can affect all bands to one site. Another site offers a different geometric path.

Frequency diversity



Bands can share one blocked path.

Attachment-path diversity

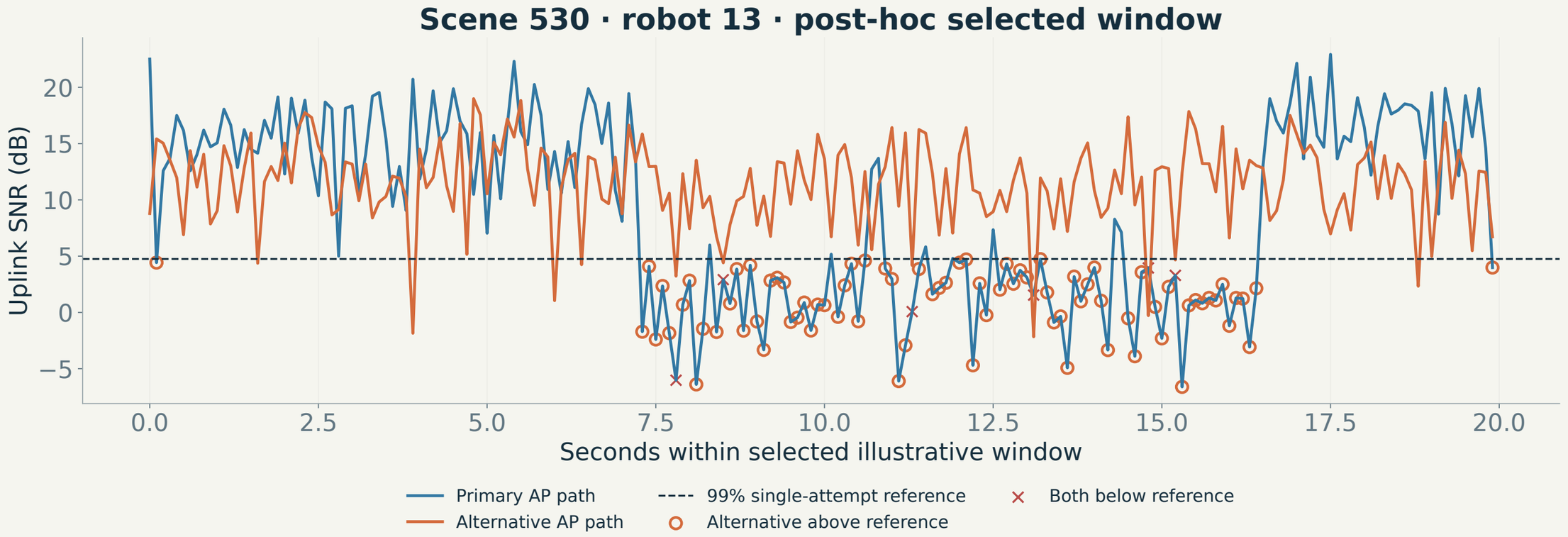


Different AP = another geometric path

INSIGHT **Cross-AP diversity is the decisive new lever in the model.**

When one AP path fades, a second path can remain usable

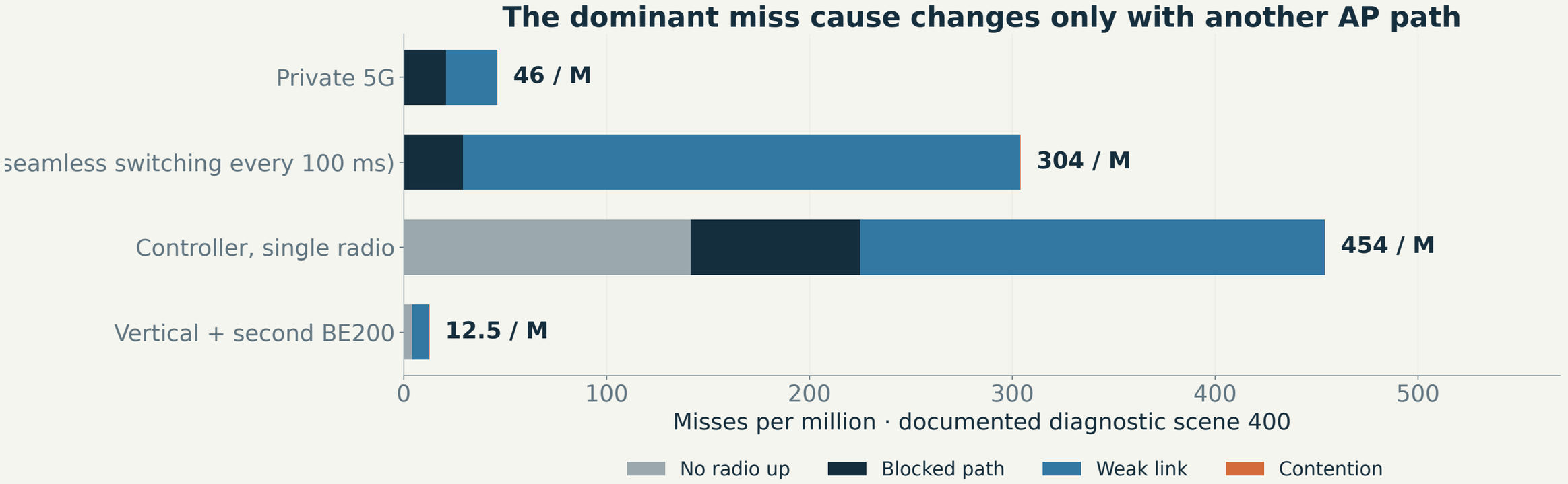
A selected 20-second trace illustrates independent-path recovery opportunities.



INSIGHT **The reference line is a 99% single-attempt operating point—not certain packet loss.**

Single-AP outages dominate the observed failure modes

The diagnostic tags misses as no radio, blocked path, weak link, contention or other.

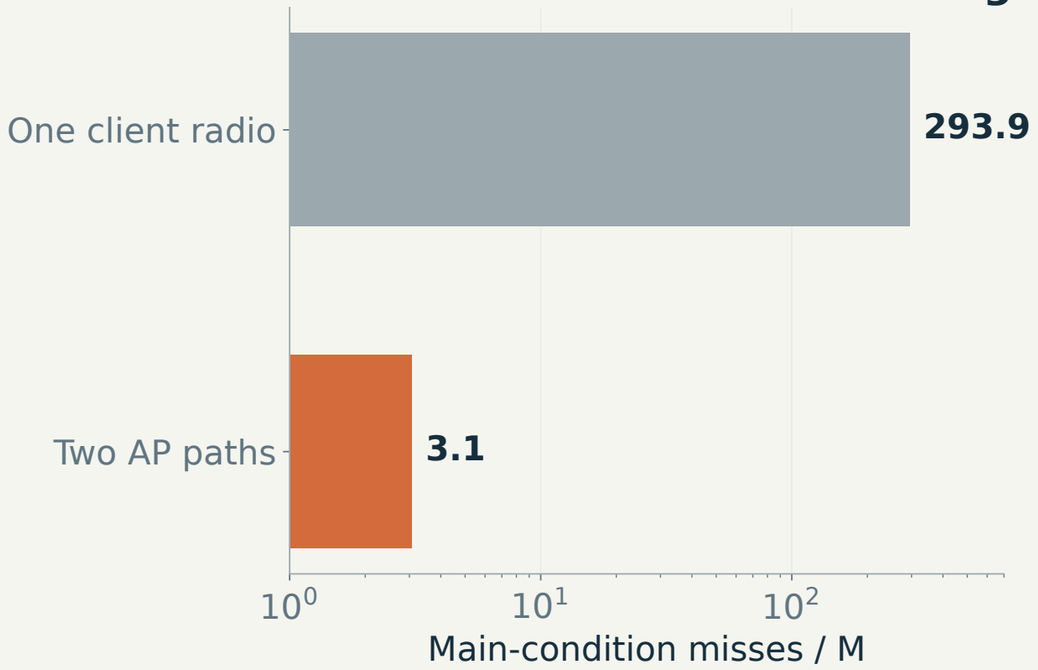


INSIGHT **Do not generalise zero contention-attributed misses to every workload.**

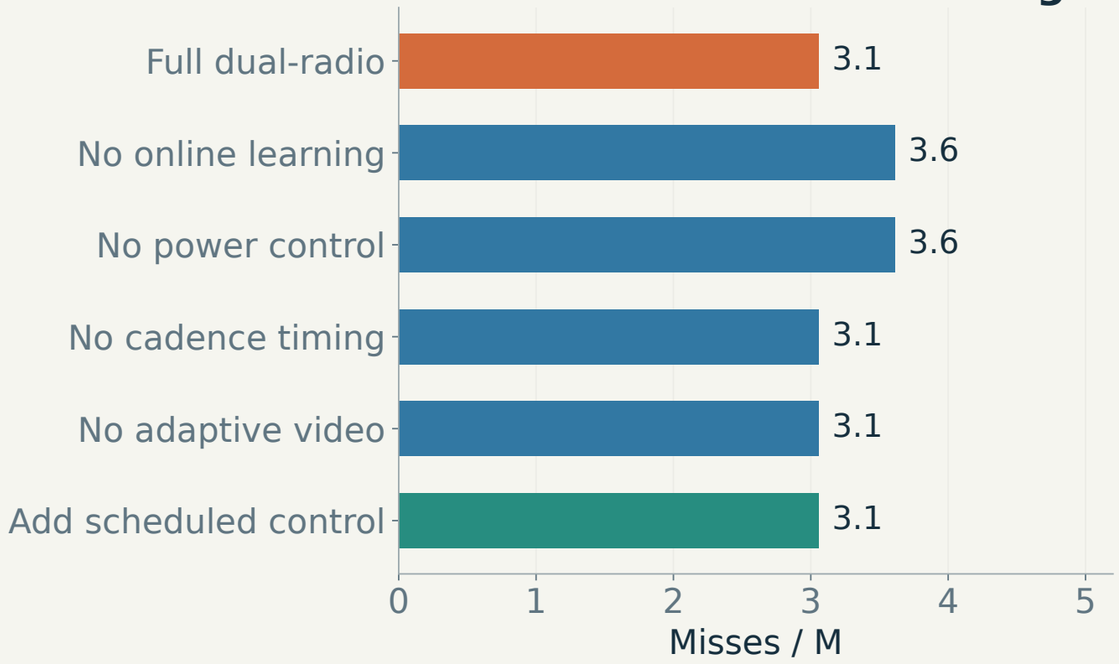
The second path matters more than the extra knobs

Online map updates, power politeness, cadence and video controls were each ablated.

The decisive architectural change



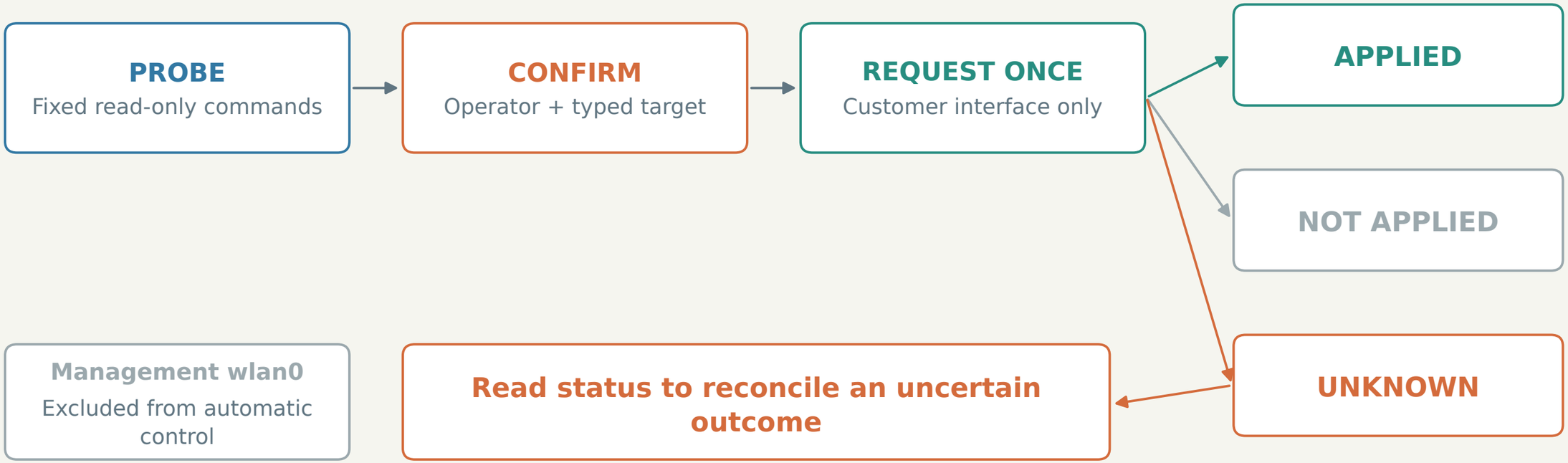
Other levers affect the margin



INSIGHT **Show every lever’s measured contribution—including no measurable effect.**

The command path makes uncertain outcomes explicit

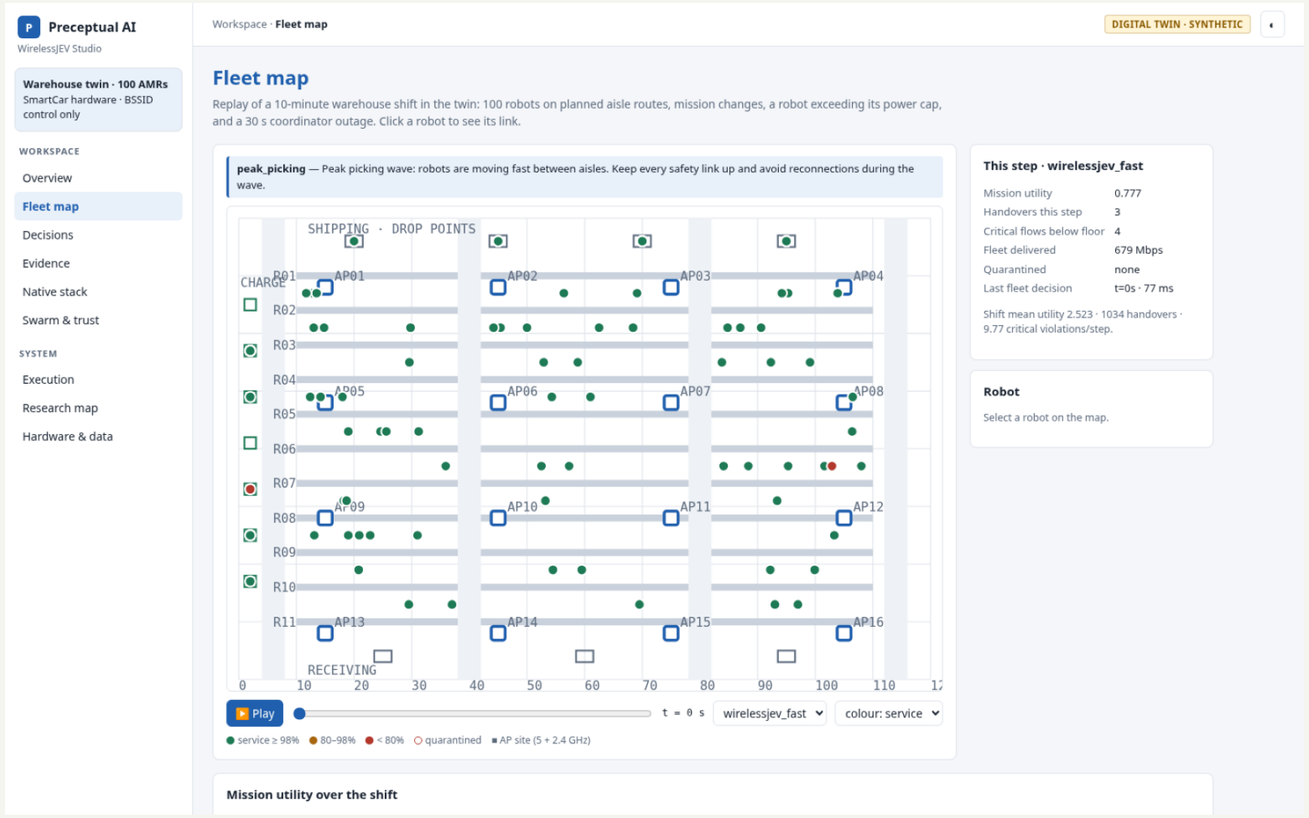
Probe first, confirm a supervised action, issue it once and reconcile by status.



INSIGHT **UNKNOWN means observe and reconcile—not automatically resend a radio-changing command.**

Fleet state, decisions and evidence are inspectable

The Studio exposes recorded robot state, link context, proposal outcomes and research evidence.



FLEET STATE

Robot position / service / link context

DECISIONS + OUTCOMES

Inspect proposal and execution records

EVIDENCE VIEWS

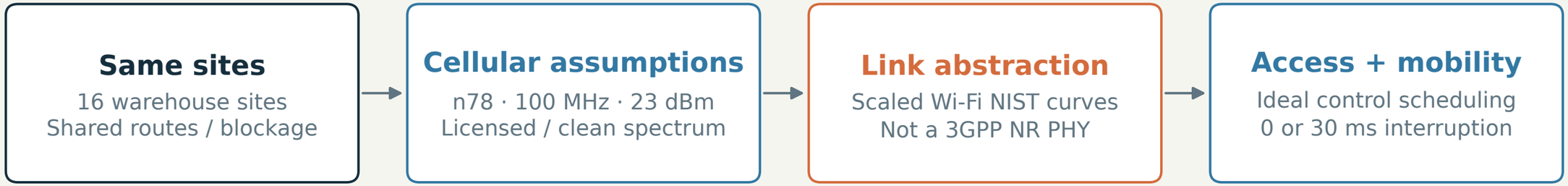
Keep simulation claims auditable

INSIGHT **Observability is a system capability—not a substitute for a live hardware trial.**

ACTUAL STUDIO SCREEN / RECORDED SIMULATION · app.js / studio_fleet_map.png

The cellular comparator is a model—not a private-5G network

Its favorable assumptions are explicit; its PHY and scheduler are still abstractions.



No inter-cell interference · no measured private network · no dual-cell packet diversity

Favorable assumptions do not make an abstract comparator a validated product benchmark.

INSIGHT **A simulated advantage does not establish commercial-5G superiority.**

Compare the same control task under the same hidden conditions

Six paired scenes, three declared interference regimes and explicit application metrics.

Same routes, fading and blockage for every compared policy.

100 robots × 300 s × 20 Hz

3.6 million commands per arm

Six paired scenes

530 531 532 533 534 535

CLEAN

λ = 0 · interference off

MAIN

λ = 0.01 · interference on

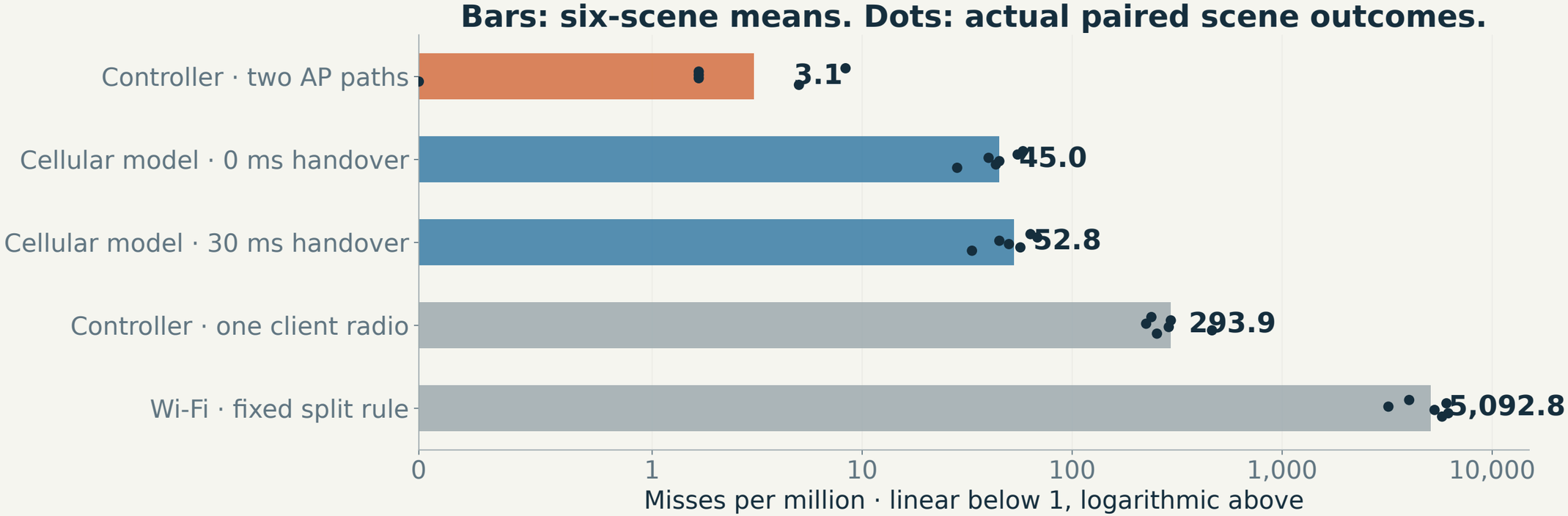
STRESS

λ = 0.03 · +10 dB at 2.4 GHz

INSIGHT **3.6 million commands per arm still come from only six independent scene realisations.**

Two AP paths reduce modelled misses by about 17×

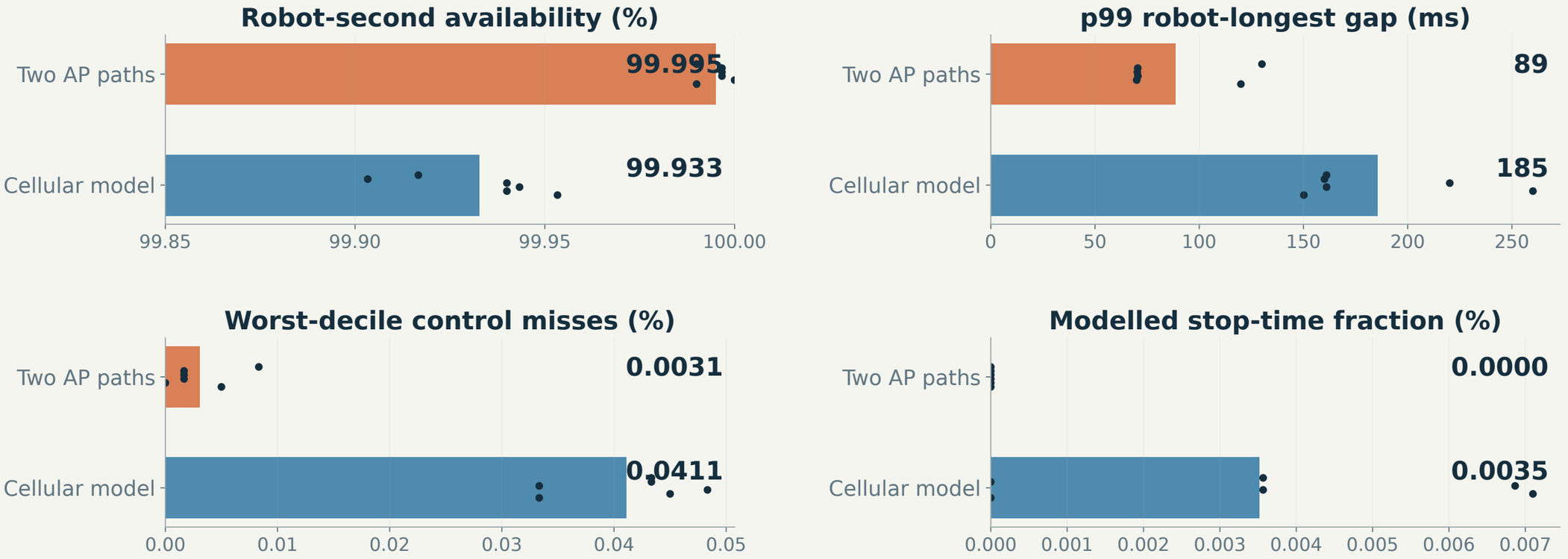
Main condition: 3.1 missed commands per million versus 52.8 for the 30 ms cellular model.



INSIGHT **The headline depends on cross-AP diversity, not existing single-radio hardware.**

The same experiment measures the tail and interruption

Availability, longest gaps and simulated stop time accompany the mean miss rate.

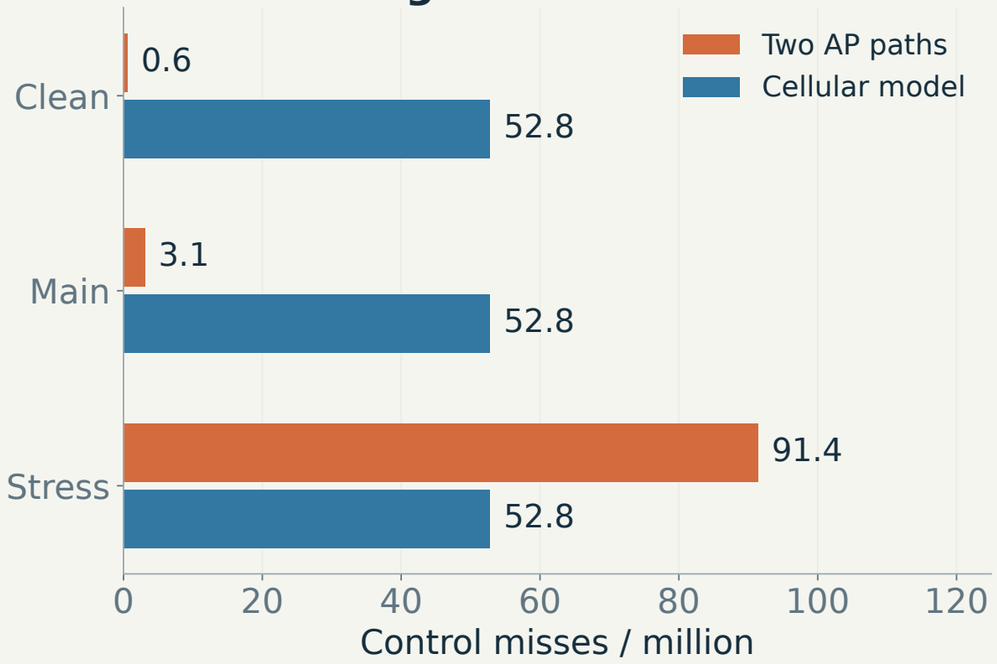


INSIGHT **These are operational proxies—not a field uptime SLA or safety certification.**

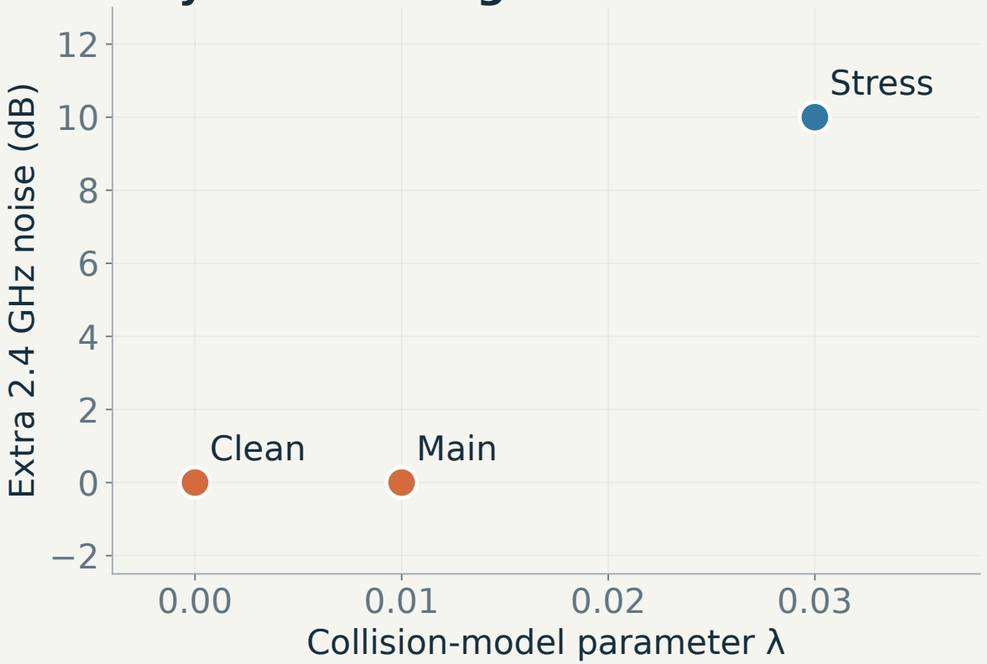
The cellular model wins in the hardest interference regime

Heavier collisions plus a 10 dB 2.4 GHz noise rise reverse the main-condition ranking.

The ranking reverses under stress



Only these configurations were tested



INSIGHT **The supported claim has an operating envelope.**

Three hardware routes can expose a second AP path

Only the added-radio path underpins the completed 100-robot comparison.

ONE BE200

Two station associations
Retune after primary retries



Support unverified

ONBOARD + BE200

Use the Pi radio as path B
Move management out of band



Weaker link; operational rework

SECOND CLIENT RADIO

Independent AP attachment
USB adapter / second card

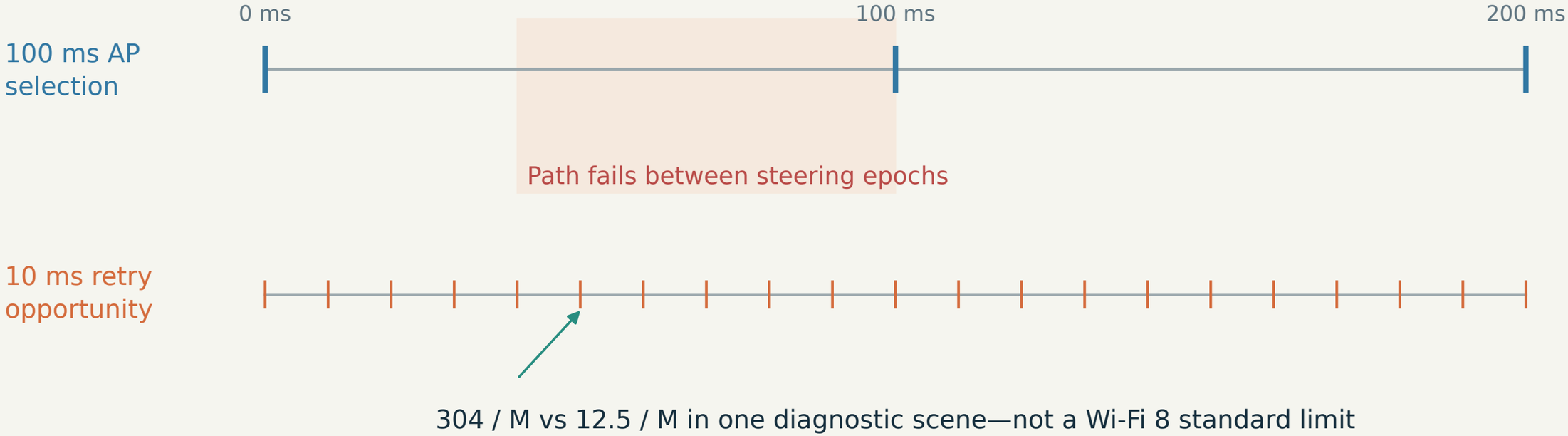


Assumed in completed main test

INSIGHT **Probe capability first; do not assume Wi-Fi 7 means two usable associations.**

Per-packet path recovery differs from 100 ms steering

The declared seamless-switching comparator chooses one AP between decision epochs.

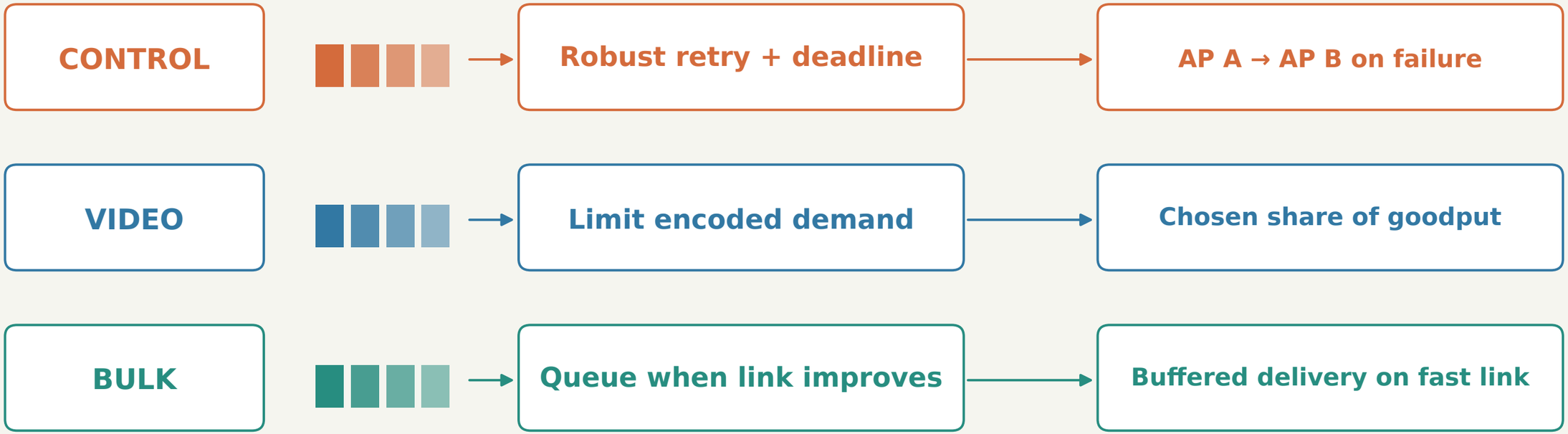


INSIGHT **This comparison tests policy granularity—not a fundamental Wi-Fi 8 standard limit.**

Control, video and bulk share a radio—but not a policy

Map traffic to suitable links, limit encoded demand and retain deferred data in a queue.

Application objectives reach the link scheduler through explicit class-specific controls.



INSIGHT **Cross-layer control connects application deadlines to link and queue decisions.**

The system covers estimation, decisions, recovery and operations

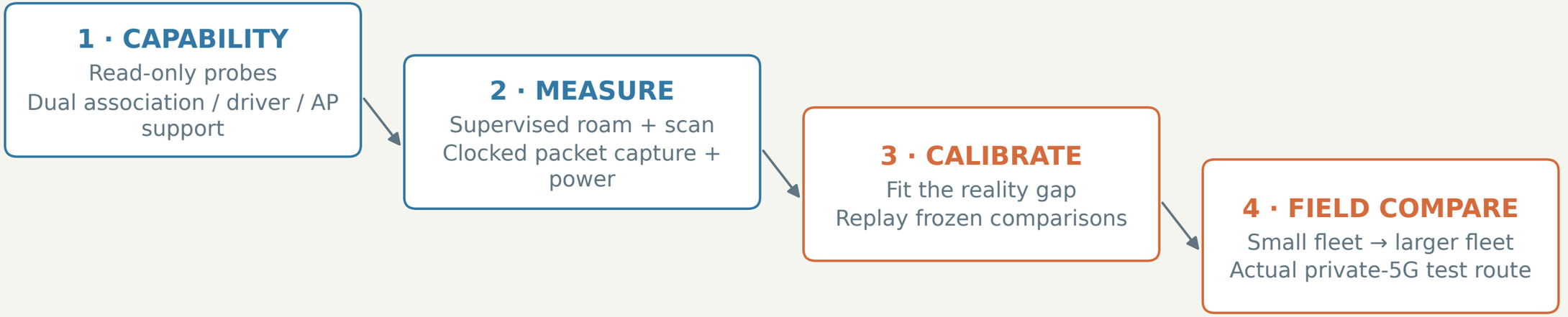
A complete feedback chain with explicit interfaces and a separate physical qualification boundary.

	Executed simulation	Separate software tools	Field qualified
Spatial map + uncertainty	●	—	—
Mission → native policy knobs	●	—	—
AP planning + trust gate	●	—	—
Control retries + cross-AP path	●	—	—
Traffic-to-link mapping	●	—	—
Video / bulk demand control	●	—	—
Delivery feedback + online map	●	—	—
Rich contract proposal interface	—	●	—
Supervised handover diagnostics	—	●	—

INSIGHT **The connected simulation is comprehensive; the field-execution boundary remains measurable.**

Four evidence gates turn the model into a field claim

Capability probe → measure → calibrate and replay → controlled fleet comparison.



No stage has yet produced physical fleet qualification.

INSIGHT **Real hardware is the next evidence source—not another favorable simulation.**

A bounded pilot needs equipment—not invented prices

Four APs, a ten-robot milestone, conditional client radios and independent measurement.

4
Wi-Fi 7 APs

10
Robot milestone

12
Conditional adapters
10 + 2 spares

2
Capture adapters
+ one laptop

2
Power meters

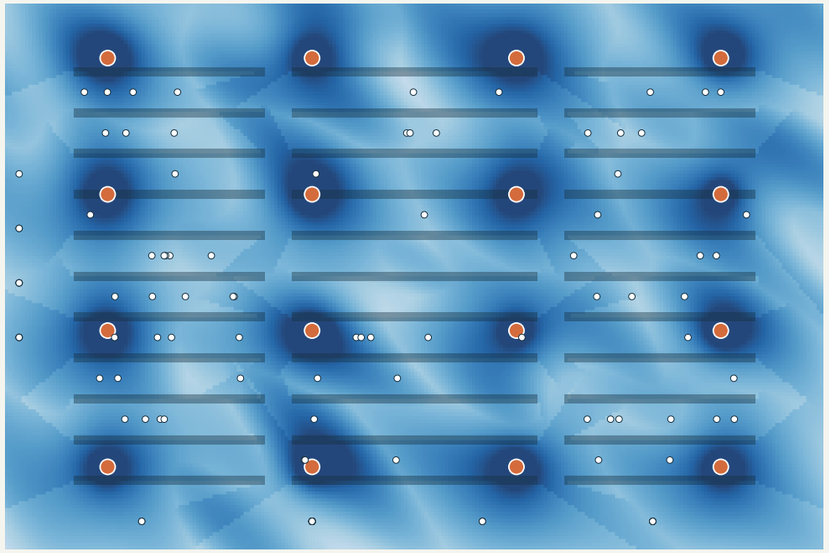
Management: 10 out-of-band paths if onboard Wi-Fi is reused

Partner lab: private-5G network + 2+ modems

INSIGHT **The second-radio bill depends on the first capability probe.**

Validate the paths, calibrate the model, then scale

The working prototype and evidence already tell us what the field trial must resolve.



LEARN
Site map + explicit uncertainty

DECIDE
Mission-driven numerical policy + gate

RECOVER
Another AP path when one fails

Next: measured interruption and a controlled pilot.

INSIGHT **Current system: RadioMap + WirelessJEV. Next milestone: measured warehouse continuity.**